

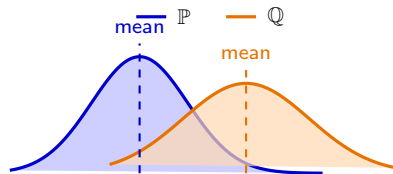
# Minimization of Maximum Mean Discrepancy (MMD)

Zonghao (Hudson) Chen

- What is Maximum mean discrepancy (MMD)?
- Minimization of MMD (miniMMD)
  - for numerical integration.
  - for sampling / generative modelling.
  - for statistical inference.
- Future work: MiniMMD for causal inference.

# Background: what is MMD?

- Given two distributions  $\mathbb{P}$  and  $\mathbb{Q}$ , how to compare them?
- Compute the difference of their means:  $|\mathbb{E}_{\mathbb{P}}[X] - \mathbb{E}_{\mathbb{Q}}[X]|$ .
- Compute the difference of their second moments:  $|\mathbb{E}_{\mathbb{P}}[X^2] - \mathbb{E}_{\mathbb{Q}}[X^2]|$ .
- The first two moments are not enough!



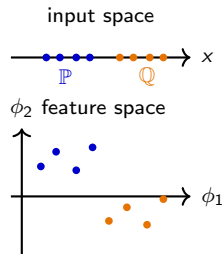
**Goal:** Compare  $\mathbb{P}$  and  $\mathbb{Q}$  with a general  $f$ , i.e.,  $|\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|$ ?

# Background: Maximum Mean Discrepancy (MMD) [GBR<sup>+</sup>12]

- $\mathcal{H}$  is a reproducing kernel Hilbert space (RKHS) with kernel  $k$ .

$$\text{MMD}(\mathbb{P}, \mathbb{Q}) := \sup_{f \in \mathcal{H}} |\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|.$$

- The RKHS contains functions that are linear combinations of the feature maps  $\phi : \mathcal{X} \rightarrow \mathcal{H}$ ,  $f = \sum_{i=1}^n \alpha_i \phi(x_i)$ .



## Theorem

*MMD admits a closed-form expression:*

$$\text{MMD}^2(\mathbb{P}, \mathbb{Q}) = \mathbb{E}_{X, X' \sim \mathbb{P}}[k(X, X')] + \mathbb{E}_{Y, Y' \sim \mathbb{Q}}[k(Y, Y')] - 2\mathbb{E}_{X \sim \mathbb{P}, Y \sim \mathbb{Q}}[k(X, Y)].$$

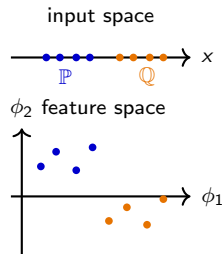
- MMD is a probability divergence ( $\text{MMD}(\mathbb{P}, \mathbb{Q}) = 0 \iff \mathbb{P} = \mathbb{Q}$ ) if  $k$  is characteristic (Gaussian) [SFL11].

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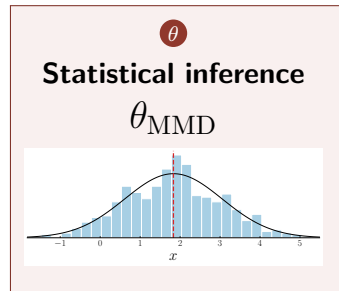
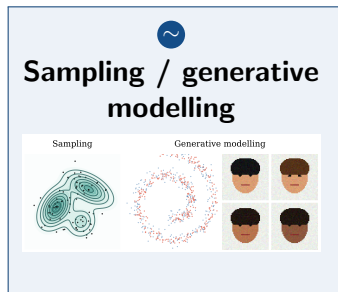
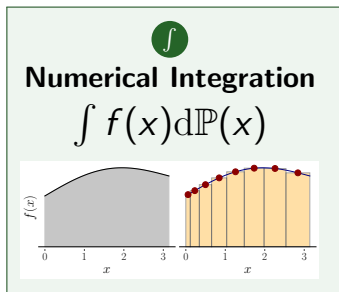
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# Background: Maximum Mean Discrepancy (MMD)

## We now have MMD

A kernel-based way to compare probability distributions  $\mathbb{P}$  and  $\mathbb{Q}$ .

## What can we use it for?



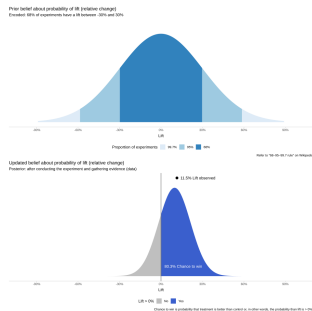
# Numerical Integration

# Part I: Numerical Integration

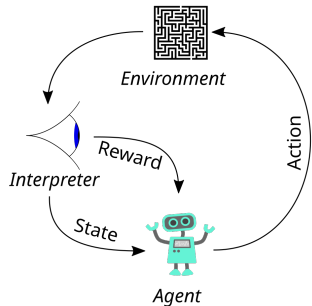
## Where do integrals appear in computer science?



**Computer graphics**  
Rendering



**Bayesian inference**  
Posterior expectations



**Reinforcement learning**  
Expected returns



# Part I: Numerical Integration

- Given a function  $f$  and a distribution  $\mathbb{P}$ , how to compute  $\int f(x)d\mathbb{P}(x)$ ?
- Given a set of points  $\{x_i\}_{i=1}^n$  and function evaluations  $\{f(x_i)\}_{i=1}^n$ .
- Quadrature rule:

$$\hat{I} = \sum_{i=1}^n w_i f(x_i).$$

- When  $w_1 = \dots = w_n = 1/n$ , this is Monte Carlo.  $\hat{I}_{MC} = \frac{1}{n} \sum_{i=1}^n f(x_i)$ .

$$\mathbb{E} \left[ |\hat{I}_{MC} - I| \right] = \mathcal{O}(n^{-1/2}).$$

- Or equivalently, to obtain an  $\epsilon$ -accurate estimate, we need  $n = \mathcal{O}(\epsilon^{-2})$  samples.

**Question:** Can we do better than  $\mathcal{O}(n^{-1/2})$ ?

**Answer:** Find better weights  $w_1, \dots, w_n$ .

# Part I: Numerical Integration

- Suppose  $f \in \mathcal{H}$  is in the RKHS associated with  $k$ .
- Denote  $\hat{\mathbb{P}} = \sum_{i=1}^n w_i \delta_{x_i}$  as the weighted empirical distribution.

$$\left| I - \hat{I} \right|^2 = \left| \int f(x) d\mathbb{P}(x) - \sum_{i=1}^n w_i f(x_i) \right|^2 \leq \|f\|_{\mathcal{H}}^2 \cdot \text{MMD}^2(\hat{\mathbb{P}}, \mathbb{P}).$$

- The right-hand side is the worst-case error of the quadrature rule.

$$w_{1:n} := \arg \min_{w_{1:n} \in \mathbb{R}^n} \text{MMD} \left( \sum_{i=1}^n w_i \delta_{x_i}, \mathbb{P} \right),$$
$$w_{1:n} = \mathbb{E}_{X \sim \mathbb{P}} [k(X, x_{1:n})] K(x_{1:n}, x_{1:n})^{-1}.$$

- The kernel quadrature (KQ) rule:  $\hat{I}_{\text{KQ}} = \sum_{i=1}^n w_i f(x_i)$ .

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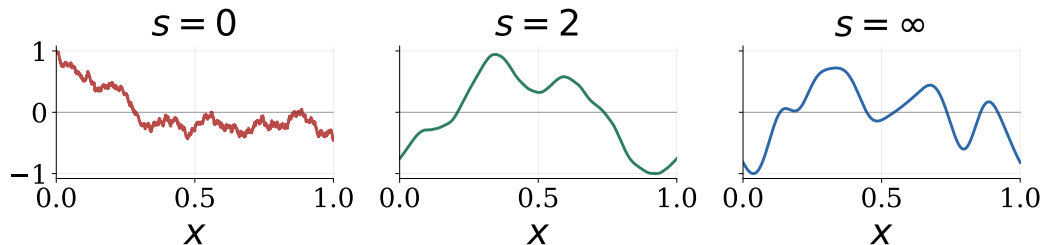
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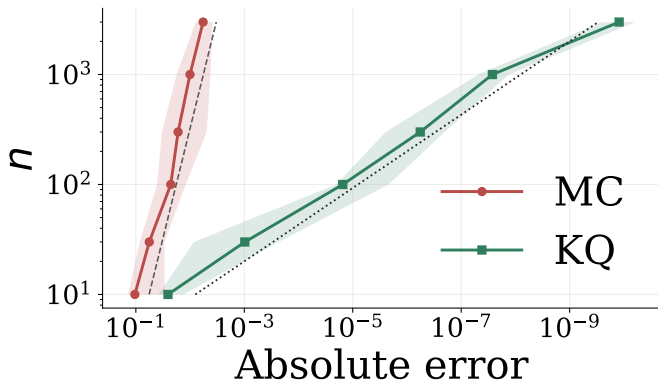
# Part I: Numerical Integration

**Rate:** KQ achieves  $\mathcal{O}(n^{-s/d})$  rate.

- $s$  is the smoothness of  $f$ ,  $d$  is the dimension of input  $x$ .
- Better than  $\mathcal{O}(n^{-1/2})$  when  $s > d/2$ .



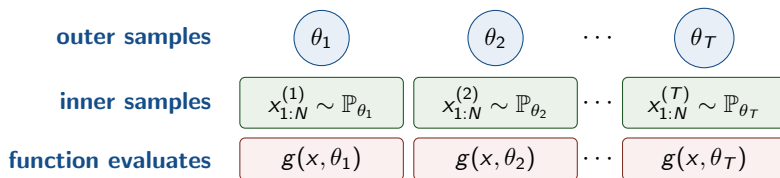
## Part I: Numerical Integration



- $s = 4$  and  $d = 1$ .

# Part I: Numerical Integration

- Nested expectation:  $I := \mathbb{E}_{\theta \sim \mathbb{Q}} [f(\mathbb{E}_{X \sim \mathbb{P}_\theta} [g(X, \theta)])]$



Total simulator evaluations:  $N \times T$

Stage I: inner KQ

$$\hat{J}_{\text{KQ}}(\theta_t) = \sum_{n=1}^N w_{n,t}^x g(x_n^{(t)}, \theta_t)$$

Stage II: outer KQ

$$\hat{I}_{\text{NKQ}} = \sum_{t=1}^T w_t^\Theta f(\hat{J}_{\text{KQ}}(\theta_t))$$

Samples to achieve error  $|I - \hat{I}| \leq \Delta$ :

NMC:  $\mathcal{O}(\Delta^{-4})$  or  $\mathcal{O}(\Delta^{-3})$

MLMC:  $\mathcal{O}(\Delta^{-2})$

NQMC:  $\mathcal{O}(\Delta^{-2.5})$

NKQ:  $\tilde{\mathcal{O}}(\Delta^{-d_X/s_X - d_\Theta/s_\Theta})$

## Conditional Bayesian Quadrature (UAI 2025)

Z. Chen, M. Naslidnyk, A. Gretton, F.-X. Briol

- Extension to conditional expectations:  $I(\theta) = \mathbb{E}_{X \sim p(\cdot|\theta)}[g(X, \theta)]$ .
- Weighted rule:  $\hat{I}_{\text{CBQ}}(\theta) := \sum_{t=1}^T \sum_{n=1}^N w_{n,t} g(x_n^{(t)}, \theta_t)$ .

## Nested Expectations with Kernel Quadrature (ICML 2025)

Z. Chen, M. Naslidnyk, F.-X. Briol

- Extension to nested expectations:  $I := \mathbb{E}_{\theta \sim \mathbb{Q}} [f(\mathbb{E}_{X \sim \mathbb{P}_\theta} [g(X, \theta)])]$ .
- Weighted rule:  $\hat{I}_{\text{NKQ}} = \sum_{t=1}^T w_t^\Theta f\left(\sum_{n=1}^N w_{n,t}^\chi g(x_n^{(t)}, \theta_t)\right)$ .

## BayesSum: Bayesian Quadrature in Discrete Spaces (ProbNum 2026)

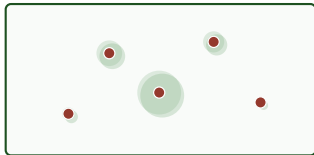
S. S. Kang, F.-X. Briol, T. Karvonen, Z. Chen

- Extension to sums:  $I = \mathbb{E}_{X \sim \mathbb{P}}[f(X)] = \sum_{x \in \mathcal{X}} f(x)p(x)$ .

# Sampling / Generative Modelling

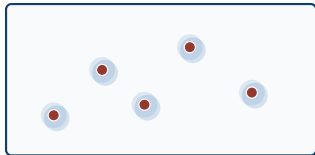
## Quadrature

fixed locations, choose weights



## Sampling / generation

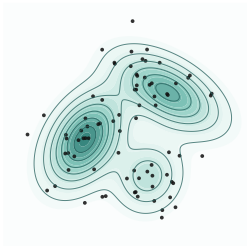
equal weights, choose locations



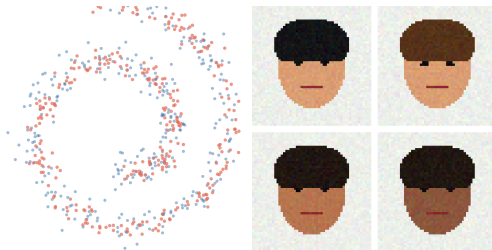


## Part II: Sampling / Generative modelling

Sampling



Generative modelling



- Given a target distribution  $\mathbb{Q}$ , how to draw samples  $\{x_i\}_{i=1}^n$  from it?
  - When  $\mathbb{Q}$  is known up to normalization, this is sampling.
  - When  $\mathbb{Q}$  is known with i.i.d samples, this is generative modelling.

**Problem:** How to learn a target probability distribution  $\mathbb{Q}$  on  $\mathbb{R}^d$ .

- This problem can be written as an **optimization** problem on  $\mathcal{P}(\mathbb{R}^d)$ , the space of probability distributions.

$$\arg \min_{\mathbb{P} \in \mathcal{P}(\mathbb{R}^d)} D(\mathbb{P}, \mathbb{Q}).$$

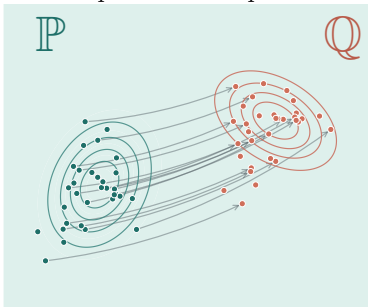
- Here  $D$  is a similarity metric or distance, e.g. Kullback–Leibler divergence / MMD.
  - $D(\mathbb{P}_1, \mathbb{P}_2) = 0$  if and only if  $\mathbb{P}_1 = \mathbb{P}_2$ .
- How to find the minimum? Gradient descent!

## Part II: Sampling / Generative modelling

**Question:** Wait... How to define gradients on  $\mathcal{P}(\mathbb{R}^d)$ ?

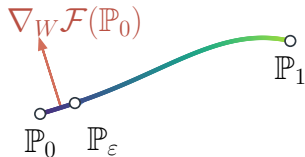
**Answer:** Wasserstein gradient ! [Ott01]

Optimal transport



Wasserstein gradient

$$\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \rightarrow \mathbb{R}$$



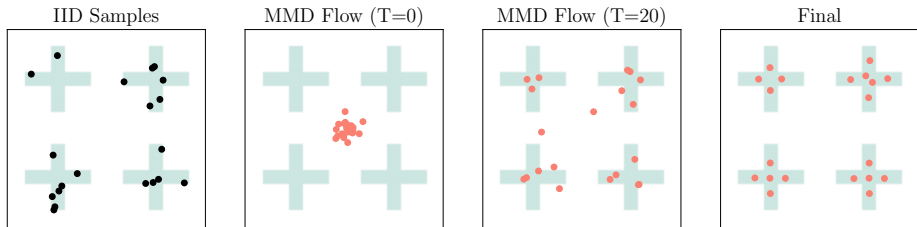
$$\nabla_W \mathcal{F}(\mathbb{P}_0) = \lim_{\varepsilon \rightarrow 0} \frac{\mathcal{F}(\mathbb{P}_\varepsilon) - \mathcal{F}(\mathbb{P}_0)}{\varepsilon}$$

## Part II: Sampling / Generative modelling

- Unsurprisingly, we use MMD as the divergence.

$$\arg \min_{\mathbb{P} \in \mathcal{P}(\mathbb{R}^d)} \text{MMD}^2(\mathbb{P}, \mathbb{Q}).$$

- A (Wasserstein) gradient descent of  $\text{MMD}^2(\mathbb{P}, \mathbb{Q})$  gives a sequence of distributions:  $\mathbb{P}_0, \mathbb{P}_1, \dots, \mathbb{P}_t$  [AKSG19].
  - In the limit,  $\mathbb{P}_t$  converges to  $\mathbb{Q}$  and  $\text{MMD}(\mathbb{P}_t, \mathbb{Q}) \rightarrow 0$  (or not?).



- MMD flow generates more **uniform** samples.

## Part II: Sampling / Generative modelling



- MMD flow does not outperform diffusion-based generative models [GBG25].
- A novel framework of generative modelling & sampling.

**Challenge:** Non-convexity of the squared MMD objective:  $\lim_{t \rightarrow \infty} \text{MMD}^2(\mathbb{P}_t, \mathbb{Q}) > 0$ .

- $\chi^2(\mu \parallel \pi)$  is convex in  $\mu$  [OT11], yet hard to implement.
- DrMMD *interpolates* between MMD ( $\lambda = \infty$ ) and  $\chi^2$  ( $\lambda = 0$ ).

$$\text{DrMMD}(\mu, \pi) = \sup_{f \in \mathcal{H}: \|f\|_{L_2(\mu)}^2 + \lambda \|f\|_{\mathcal{H}}^2 \leq 1} \int f \, d(\mu - \pi).$$

- DrMMD flow converges in KL divergence:  $\lim_{t \rightarrow \infty} \text{KL}(\mathbb{P}_t, \mathbb{Q}) = 0$ .
  - When  $\pi$  is ‘Gaussian-like’.

### (De)-regularized MMD Gradient Flow (JMLR 2025)

Z. Chen, A. Mustafi, P. Glaser, A. Korba, A. Gretton, B. K. Sriperumbudur

- DrMMD interpolates between MMD and  $\chi^2$  divergence.
- Convergence in KL divergence:  $\lim_{t \rightarrow \infty} \text{KL}(\mathbb{P}_t, \mathbb{Q}) = 0$ .

### Sobolev Regularized MMD Gradient Flow (Arxiv)

C. Tian, B. K. Sriperumbudur, A. Gretton, Z. Chen

- Gradient regularization on the critic [MSR19].

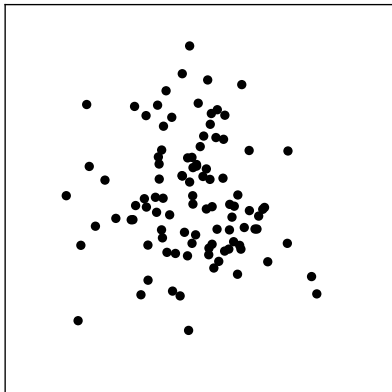
$$\text{SrMMD}(\mu, \pi) = \sup_{f \in \mathcal{H}: \|\nabla f\|_{L_2^d(\mu)}^2 + \lambda \|f\|_{\mathcal{H}}^2 \leq 1} \int f \, d(\mu - \pi).$$

- Convergence in MMD:  $\lim_{t \rightarrow \infty} \text{MMD}(\mathbb{P}_t, \mathbb{Q}) = 0$ .
- No conditions on  $\pi$ .

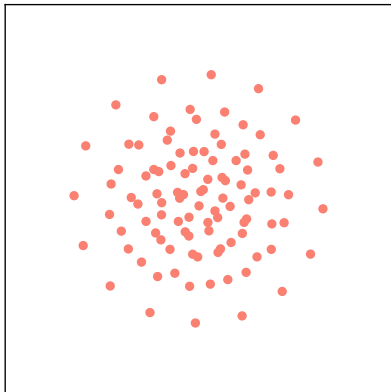
### Empirical observation:

(Wasserstein) gradient flows give samples more **uniform** than i.i.d samples [XKS22].

IID



MMD flow





- For any  $i \in \{1, \dots, n\}$ :

$$\nabla_{x_i} \text{MMD}^2 \left( \mu, \frac{1}{n} \sum_{j=1}^n \delta_{x_j} \right) = 0 \implies \frac{1}{n} \sum_{j=1}^n \nabla_1 k(x_i, x_j) - \int \nabla_1 k(x_i, y) d\mu(y) = 0.$$

- Exact integration on  $x \mapsto \nabla_1 k(x_i, x)$ .

### Exact integration subspace:

$$\mathcal{G}_n := \text{span} \left\{ \nabla_1 k(x_i, \cdot) : i \in \{1, \dots, n\} \right\} \subset \mathcal{H}.$$

- As  $n \rightarrow \infty$ ,  $\mathcal{G}_n$  approximates the whole RKHS  $\mathcal{H}$  up to constant functions.
- Stationary MMD points achieve  $\mathcal{O}(n^{-1/2})$  rate for numerical integration.

### Stationary MMD Points (ICML 2026)

Z. Chen, T. Karvonen, H. Kanagawa, F.-X. Briol, C. J. Oates

- Samples from MMD flow are better for numerical integration than IID samples.
- IID samples:  $\mathcal{O}(n^{-1/2})$ , stationary MMD points:  $o(n^{-1/2})$ .

### Thinned Mean Field Langevin Dynamics (ICML 2026)

Z. Chen, H. Kanagawa, F.-X. Briol, C. J. Oates, L. Mackey

- Computational cost of MMD flow is  $\mathcal{O}(n^2)$  due to pair-wise interaction.
- Reduce cost to  $\mathcal{O}(n^{1.5})$  without sacrificing the sample quality.
- Kernel thinning!

# Statistical Inference

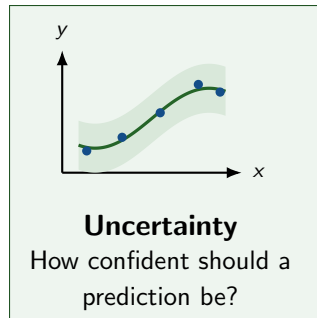
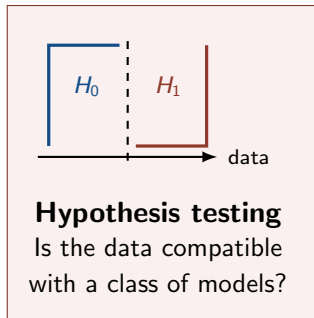
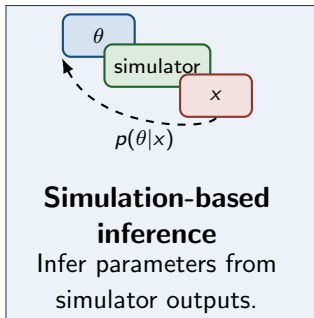
## MMD flow

$$\arg \min_{\mathbb{P} \in \mathcal{P}(\mathbb{R}^d)} \text{MMD}^2(\mathbb{P}, \mathbb{Q})$$

## Minimum MMD inference

$$\theta_{\text{MMD}} = \arg \min_{\theta \in \Theta} \text{MMD}^2(\mathbb{P}_{\theta}, \mathbb{Q})$$

## Where does statistical inference appear in computer science?



## Part III: Statistical inference

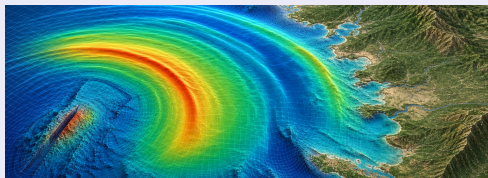
- Given a parametric model  $\{\mathbb{P}_\theta\}_{\theta \in \Theta \subseteq \mathbb{R}^p}$ .
- Infer the parameter  $\theta$  that best explains an unknown data distribution  $\mathbb{Q}$ .
- Maximum likelihood estimation is the most popular:

$$\theta_{\text{MLE}} = \arg \max_{\theta \in \Theta} \mathbb{E}_{X \sim \mathbb{Q}}[\log p_\theta(X)].$$

### What is not captured by maximum likelihood?

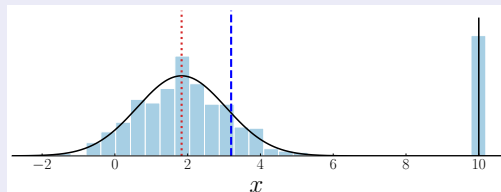
#### Likelihood-free simulators

When  $\mathbb{P}_\theta$  is a scientific simulator: we can sample  $x \sim p_\theta(x)$ , but cannot evaluate the density  $p_\theta(x)$ .



#### Robustness

A few contaminated observations can strongly move the fitted parameter.



## Part III: Statistical inference

- Minimum MMD estimator [? ? ]

$$\begin{aligned}\theta_{\text{MMD}} &= \arg \min_{\theta \in \Theta} \text{MMD}^2(\mathbb{P}_\theta, \mathbb{Q}) \\ &= \arg \min_{\theta \in \Theta} \left\{ \mathbb{E}_{x, x' \sim \mathbb{P}_\theta} [k(x, x')] - 2\mathbb{E}_{x \sim \mathbb{P}_\theta, y \sim \mathbb{Q}} [k(x, y)] + \text{const} \right\}.\end{aligned}$$

**Challenge 1:** Likelihood-free inference?

✓ Only requires samples from  $\mathbb{P}_\theta$  and  $\mathbb{Q}$ .

**Challenge 2:** Robustness to outliers?

✓ Bounded effects under bounded kernels.

**Question:** How to reliably find  $\theta_{\text{MMD}}$ ?

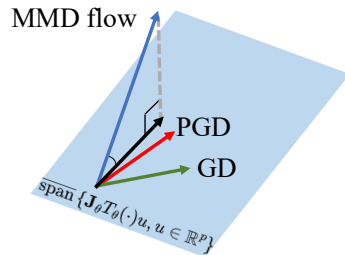
- MMD flow:

$$\mathbb{P}_{t+1} = (\text{Id} - \gamma \nabla f_{\mathbb{P}_t, \mathbb{Q}})_{\#} \mathbb{P}_t.$$

- MMD flow converges globally.
  - Adaptive lengthscale, DrMMD, SrMMD, etc.
- Consider  $\mathbb{P}_{\theta} = (T_{\theta})_{\#} \rho$ :

$$\mathbb{P}_{\theta_{t+1}} = (T_{\theta_t - \gamma \Delta \theta_t})_{\#} \rho.$$

- Choose  $\mathbb{P}_{\theta_{t+1}}$  closest to the MMD flow iterate  $\mathbb{P}_{t+1}$ .
- $\lim_{t \rightarrow \infty} \text{MMD}(\mathbb{P}_{\theta_t}, \mathbb{Q}) = \min_{\theta} \text{MMD}(\mathbb{P}_{\theta}, \mathbb{Q})$ .





### On Optimization of Minimum MMD Estimators ([Arxiv](#))

S. S. Kang, L. Sharrock, X. Cheng, F.-X. Briol, [Z. Chen](#)

- A preconditioned method / natural gradient type method.
- Adaptive kernel lengthscale.

**What next: causal inference.**

## Part IV: What next?

### Minimum MMD for counterfactual inference

**Question:** What would have happened under a different intervention?

- Treatment  $A$ .
- Counterfactual outcome:  $Y(a)$  under treatment  $A = a$ .
- Missing data/inference problem.



#### Conformal Counterfactual Inference under Hidden Confounding (KDD 2024)

[Z. Chen](#), R. Guo, J.-F. Ton, Y. Liu

#### Neural Networks for NPIV: Optimization and Generalization (Arxiv)

[Z. Chen](#), A. Nitanda, A. Gretton, T. Suzuki

#### NPIV with Observed Covariates (Arxiv)

Z. Shen\*, [Z. Chen](#)\*, D. Meunier, I. Steinwart, A. Gretton, Z. Li

# Summary

# Summary

- MMD is a powerful tool for comparing probability distributions:

$$\text{MMD}(\mathbb{P}, \mathbb{Q}) := \sup_{f \in \mathcal{H}} |\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|.$$

- Minimization of MMD in ....



## Numerical Integration

$$\min_{w_{1:n} \in \mathbb{R}^n} \text{MMD} \left( \sum_{i=1}^n w_i \delta_{x_i}, \mathbb{P} \right)$$



## Sampling / generative modelling

$$\min_{\mathbb{P} \in \mathcal{P}(\mathbb{R}^d)} \text{MMD}^2(\mathbb{P}, \mathbb{Q})$$



## Statistical inference

$$\min_{\theta \in \Theta} \text{MMD}^2(\mathbb{P}_{\theta}, \mathbb{Q})$$

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